Spoke-to-Spoke Lab — Validation Report

Summary

This report validates the spoke-to-spoke traffic hairpinning problem through a VPN gateway and evaluates two fix approaches:

1. **Direct Peering** — Remove forced tunneling UDR, disable gateway transit, peer DBX spoke directly to PE spoke ✅ *(matches real-world fix)*
2. **Adjacent Private Endpoint** — Place private endpoints in the consumer's VNet so traffic stays local ✅

Test environment: Azure hub-and-spoke lab with dedicated PE VNet in `rg-spoke-to-spoke-lab` (centralus)
Test workload: 1 GB file upload/download cycles between `vm-dbrx` (DBX spoke) and ADLS Gen2 private endpoint (PE spoke) using azcopy
Test duration: 15 minutes per configuration, 5-minute gaps between tests
Date: 2026-04-06 (all times CDT)

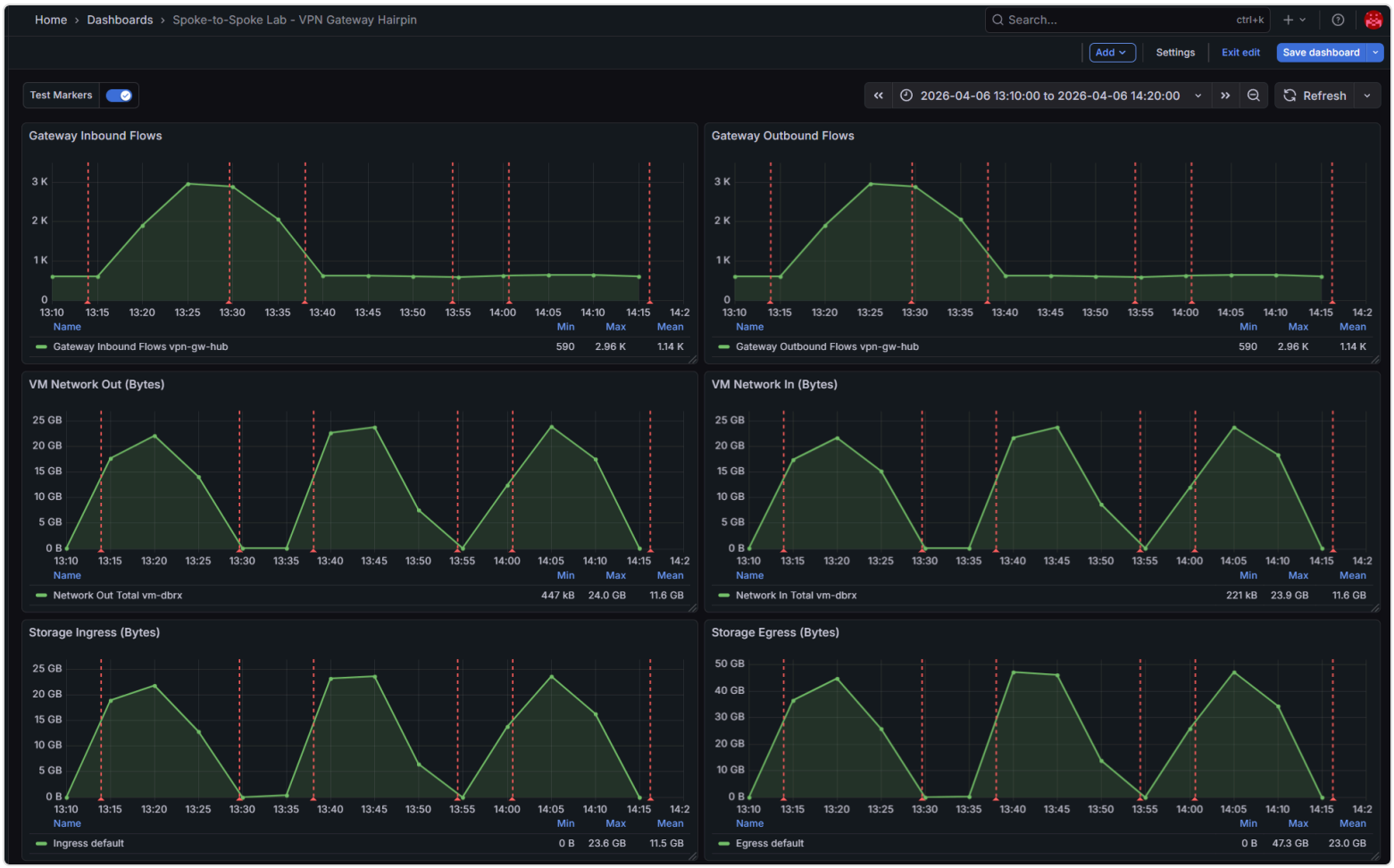
Dashboard Overview — All Three Test Runs

The screenshot below shows the complete Grafana dashboard across all three test configurations in a single view. Each vertical dashed line is a **Grafana annotation** marking a test boundary (begin or end), posted programmatically via the Grafana API using `scripts/grafana-annotate.ps1`. The annotations create clear visual markers that correlate events across all six panels simultaneously.

Reading left to right, the three traffic bursts correspond to:

1. **Broken state** (~18:14–18:29 UTC) — Gateway flows spike to ~3K, all traffic hairpins
2. **Direct Peering** (~18:38–18:54 UTC) — Gateway flat at baseline, traffic flows peer-to-peer
3. **Adjacent PE** (~19:00–19:16 UTC) — Gateway remains flat, storage data stays in consumer spoke

Each panel includes **min, max, and mean** values in the legend table for quantitative comparison. The gateway panels use a **fixed Y-axis (0–3,500)** so the flat baseline in Fix 2 is visually distinct from the broken state's spike — without fixed scaling, auto-scale would make the ~600 ancillary flows appear as volatile spikes.



The following sections break down each configuration in detail.

The Problem: VPN Gateway Hairpin

> **Note on gateway type**: This lab uses a VPN gateway (VpnGw1), but the same hairpin behavior occurs with an **ExpressRoute gateway**. Both gateway types support `allowGatewayTransit` / `useRemoteGateways` on VNet peerings, and both will process spoke-to-spoke traffic when a catch-all UDR forces `0.0.0.0/0 → VirtualNetworkGateway`. The fixes demonstrated here — direct peering and adjacent private endpoints — apply equally to ExpressRoute environments.

Real-World Scenario

In the customer environment that motivated this lab, the topology includes a **dedicated Private Endpoint VNet** — a separate spoke that hosts all private endpoints centrally, rather than co-locating PEs within each service's spoke. The Databricks workloads in their own spoke(s) access ADLS Gen2 through PEs in this dedicated VNet:

```
Databricks Spoke(s)          Hub                 PE VNet            ADLS

┌──────────────┐        ┌──────────┐        ┌──────────┐       ┌──────────┐
│  DBX VMs     │        │ ER/VPN   │        │ PE (DFS) │       │ Storage  │
│              │──────→ │ Gateway  │──────→ │ PE (Blob)│──────→ │ Account  │
│              │        │          │        │          │       │          │
```

```
└──────────────┘        └──────────────┘        └──────────────┘        └──────────────┘

      peering ──► Gateway Transit ──► peering
                  ALL traffic hairpins
```

The fix applied was **direct VNet peering between the Databricks spokes and the PE VNet**, eliminating the gateway from the data path. This maps directly to **Fix 1 (Direct Peering)** in this lab.

Lab Architecture

This lab reproduces the customer's three-VNet topology:

- **Spoke 1** (`vnet-spoke-dbrx` , 10.101.0.0/16) — Databricks simulation with Linux VM
- **Hub** (`vnet-hub` , 10.100.0.0/16) — VPN Gateway in GatewaySubnet
- **PE Spoke** (`vnet-spoke-pe` , 10.103.0.0/16) — Dedicated PE VNet with DFS and Blob private endpoints
- **ADLS Spoke** (`vnet-spoke-adls` , 10.102.0.0/16) — Storage account (no PEs in this VNet)

Architecture (Broken State)

```
 Spoke 1 (vm-dbrx)       Hub (vnet-hub)       PE Spoke              ADLS Spoke
10.101.0.0/16            10.100.0.0/16        10.103.0.0/16         10.102.0.0/16
     │                        │                    │                      │
     │  UDR: 0/0 → GW         │                    │  PE (DFS)            │  Storage
     │                        │                    │  PE (Blob)           │  Account
     │                        │                    │                      │
     └── peering ──► VPN Gateway ◄── peering ──────┘                      │
          (useRemoteGw)   │        (useRemoteGw)                          │
                          │                                               │
                ALL spoke-to-spoke          peering ──────────────────────┘
                traffic hairpins here       (useRemoteGw)
```

All three spokes peer to the hub with gateway transit. UDRs on all spokes force `0.0.0.0/0 → VirtualNetworkGateway` . Traffic from vm-dbrx to the PE VNet (10.103.2.x) is caught by the UDR and hairpins through the VPN gateway.

Effective Routes (Broken)

```
 Source    State    Address Prefix    Next Hop Type
────────  ───────  ────────────────  ──────────────────────
Default   Active   10.101.0.0/16     VnetLocal
Default   Active   10.100.0.0/16     VNetPeering
User      Active   0.0.0.0/0         VirtualNetworkGateway   ◄── Forces ALL traffic through gateway
User      Active   68.47.19.27/32    Internet
Default   Invalid  0.0.0.0/0         Internet                ◄── Overridden by UDR
```

Key observation: No explicit routes for `10.102.0.0/16` (ADLS spoke) or `10.103.0.0/16` (PE spoke). Traffic to the PE VNet private endpoints (`10.103.2.4`) falls under the `0.0.0.0/0 → VirtualNetworkGateway` catch-all, hairpinning through the gateway.

Grafana — Broken State (13:14–13:29 CDT)

Metric	Min	Max	Mean
Gateway Inbound Flows	607	2,956	1,821
Gateway Outbound Flows	607	2,956	1,821
VM Network Out	86 kB	4.7 GB	3.6 GB
VM Network In	44 kB	5.4 GB	3.6 GB

Gateway flows ramped from **607 to 2,956** — all storage traffic hairpins through the VPN gateway.

Fix 1: Direct Spoke-to-Spoke Peering

Changes Applied

Change	Before (Broken)	After (Direct Peering)
Route tables	`0.0.0.0/0 → VirtualNetworkGateway`	Route **removed**
Hub→Spoke peering	`allowGatewayTransit: true`	`allowGatewayTransit: false`
Spoke→Hub peering	`useRemoteGateways: true`	`useRemoteGateways: false`
Spoke↔PE peering	None	**Direct peering** vnet-spoke-dbrx ↔ vnet-spoke-pe

Architecture (Direct Peering)

```
 Spoke 1 (vm-dbrx)       Hub (vnet-hub)       PE Spoke              ADLS Spoke
10.101.0.0/16            10.100.0.0/16        10.103.0.0/16         10.102.0.0/16
     │                        │                    │                      │
     │  No default UDR        │                    │  PE (DFS)            │
     │                        │                    │  PE (Blob)           │
     ├── peering ─────────────┤── peering ─────────┤                      │
     │  (no gateway transit)  │  (no gw transit)   │                      │
     │                                             │                      │
     └──────── direct peering ─────────────────────┘                      │
               Traffic goes HERE now                peering ──────────────┘
               (bypasses gateway)                   (no gw transit)
```

Effective Routes (Direct Peering)

```
 Source    State    Address Prefix    Next Hop Type
────────  ───────  ─────────────────  ──────────────────────
Default   Active   10.101.0.0/16     VnetLocal
Default   Active   10.100.0.0/16     VNetPeering
Default   Active   10.103.0.0/16     VNetPeering            ◄── Direct route to PE spoke
Default   Active   0.0.0.0/0         Internet               ◄── Default (no forced tunneling)
User      Active   68.47.19.27/32    Internet
Default   Active   10.103.2.4/32     InterfaceEndpoint      ◄── PE routes via peering
Default   Active   10.103.2.5/32     InterfaceEndpoint
```

Grafana — Direct Peering (13:38–13:54 CDT)

Metric	Min	Max	Mean
Gateway Inbound Flows	595	627	613
Gateway Outbound Flows	595	627	613
VM Network Out	116 kB	5.0 GB	3.6 GB
VM Network In	58 kB	5.4 GB	3.6 GB

Gateway flows **flat at ~600** (baseline) throughout the test. No spike — traffic flows directly via the DBX↔PE peering, completely bypassing the gateway.

Fix 2: Adjacent Private Endpoint

Concept

Place private endpoints for the storage account **in the consumer's VNet** (vnet-spoke-dbrx). The VM connects to a local PE IP (`10.101.2.x`) instead of the remote PE in the PE spoke (`10.103.2.x`). Azure creates `/32 InterfaceEndpoint` routes that are more specific than the `0.0.0.0/0` UDR, completely bypassing the forced tunneling path for storage data.

Key advantage: No changes to route tables, peering, or gateway transit. The existing "broken" routing remains intact, but storage traffic stays local.

Note: The catch-all UDR (`0.0.0.0/0 → VirtualNetworkGateway`) remains active, so ancillary traffic (AAD authentication, DNS, etc.) still transits the gateway. Only storage data is redirected via the `/32` routes.

Changes Applied

Change	Before (Broken)	After (Adjacent PE)
Route tables	`0.0.0.0/0 → VirtualNetworkGateway`	**Unchanged**
Hub↔Spoke peering	`allowGatewayTransit: true`	**Unchanged**
Spoke→Hub peering	`useRemoteGateways: true`	**Unchanged**

Change	Before (Broken)	After (Adjacent PE)
vnet-spoke-dbrx subnets	`subnet-dbrx` only	Added `subnet-pe` (10.101.2.0/24)
Private endpoints	DFS + Blob PEs in PE spoke only	**Added** DFS + Blob PEs in spoke-dbrx

Architecture (Adjacent PE)

```
 Spoke 1 (vm-dbrx)       Hub (vnet-hub)       PE Spoke              ADLS Spoke
10.101.0.0/16            10.100.0.0/16        10.103.0.0/16         10.102.0.0/16
     │                        │                    │                     │
     │  UDR: 0/0 → GW         │                    │  PE (DFS)           │  Storage
     │  (STILL ACTIVE)        │                    │  PE (Blob)          │  Account
     │                        │                    │                     │
     ├── peering ──────► VPN Gateway ◄── peering ──┘                     │
     │                                             peering ──────────────┘
     │  ┌─────────────────┐
     │  │ subnet-pe       │
     │  │ 10.101.2.0/24   │
     │  │ pe-dfs-local  ─────── Azure backbone ──────────► ADLS
     │  │ pe-blob-local ─────── Azure backbone ──────────► ADLS
     │  └─────────────────┘
     │
     └── VM traffic goes to LOCAL PE (bypasses gateway + PE spoke)
```

Effective Routes (Adjacent PE)

```
 Source    State    Address Prefix    Next Hop Type
────────  ───────  ─────────────────  ──────────────────────────
Default   Active   10.101.0.0/16      VnetLocal
Default   Active   10.100.0.0/16      VNetPeering
User      Active   0.0.0.0/0          VirtualNetworkGateway    ◄── UDR STILL ACTIVE
User      Active   68.47.19.27/32     Internet
Default   Invalid  0.0.0.0/0          Internet
Default   Active   10.101.2.4/32      InterfaceEndpoint        ◄── Local DFS PE (/32 overrides UDR)
Default   Active   10.101.2.5/32      InterfaceEndpoint        ◄── Local Blob PE (/32 overrides UDR)
```

Key observation: The `/32 InterfaceEndpoint` routes at `10.101.2.4` and `10.101.2.5` override the catch-all UDR. Storage traffic stays within vnet-spoke-dbrx.

Grafana — Adjacent PE (14:00–14:16 CDT)

Metric	Min	Max	Mean
Gateway Inbound Flows	610	641	627
Gateway Outbound Flows	610	641	627
VM Network Out	84 kB	5.1 GB	3.6 GB
VM Network In	43 kB	5.4 GB	3.6 GB

Gateway flows remained **essentially flat at 610–641** — the gateway is not processing storage data. The slight variation (~30 flow range) is ancillary traffic (AAD auth, DNS queries) still caught by the catch-all UDR.

Conclusion

Comparison of All Configurations

Metric	Broken (Hairpin)	Fix 1 (Direct Peering)	Fix 2 (Adjacent PE)
Status	⚠️ Working (inefficient)	✅ Working	✅ Working
Gateway Flows (Max)	**2,956**	627 (baseline)	**641** (flat baseline)
Gateway in data path?	Yes (all traffic)	**No**	**No** (storage data bypassed)
VM Network Out (Max)	4.7 GB	5.0 GB	5.1 GB
VM Network In (Max)	5.4 GB	5.4 GB	5.4 GB
Routing changes	—	UDR removed, gateway transit disabled	**None**
Peering changes	—	DBX↔PE spoke direct peering added	**None**
Infrastructure added	—	None	PE subnet + 2 PEs

Fix 1: Direct Peering (Recommended)

- **Matches the real-world customer fix** — the Databricks team peered their spoke directly to the PE spoke

- Removes the gateway from the data path entirely by fixing the routing architecture
- Requires changes to route tables, peering settings, and adding spoke-to-spoke peering
- Best for clean network architecture and when spoke-to-spoke direct communication is feasible

Fix 2: Adjacent Private Endpoint

- Bypasses the gateway without changing any routing or peering settings
- The forced tunneling UDR remains active, but `/32 InterfaceEndpoint` routes override it
- Minimally invasive — only adds a PE subnet and two private endpoints in the consumer's VNet
- Valid alternative when direct peering between spokes isn't feasible or when preserving all non-PE traffic inspection is required
- **Note**: Ancillary traffic (AAD auth, DNS, etc.) still transits the gateway due to the catch-all UDR — only storage data is bypassed

Other Approaches (Not Tested)

Azure Virtual WAN (VWAN) would also address this issue. VWAN's hub provides native spoke-to-spoke routing without requiring manual peering or UDRs — the VWAN hub router automatically learns spoke prefixes and forwards traffic directly between spokes. In a VWAN topology, spoke-to-spoke traffic does not hairpin through the VPN or ExpressRoute gateway; it is routed through the VWAN hub router at no additional gateway cost. VWAN also supports routing intent and policies that provide more granular control over inter-spoke traffic flows. However, VWAN requires migrating from a traditional hub-and-spoke architecture and was not tested in this lab.

Security and Compliance Implications

Each fix has different security posture trade-offs. Organizations subject to regulatory controls (e.g., PCI-DSS, HIPAA, FedRAMP) should evaluate these carefully.

Broken State (Baseline)

- **Forced tunneling**: All traffic (`0.0.0.0/0`) routes through the gateway, providing a **central inspection point** — this is often a compliance requirement for environments mandating egress filtering or network-level DLP
- **Gateway as chokepoint**: The gateway sees all flows, enabling logging, IDS/IPS integration, and traffic analytics
- **Risk**: Gateway saturation degrades availability — PPS and throughput limits can cause packet drops, impacting SLA compliance

Fix 1: Direct Peering

- **Removes forced tunneling entirely** — spoke-to-spoke and internet traffic no longer transits the gateway
- **No central inspection point**: Traffic between spokes bypasses the hub completely. If your compliance framework requires all inter-VNet traffic to pass through a firewall or NVA for inspection, this fix **violates that requirement**
- **Mitigation**: Deploy an Azure Firewall or NVA in the hub and route spoke-to-spoke traffic through it instead of the gateway. This was not implemented in this lab.
- **NSG enforcement**: Without forced tunneling, NSGs on spoke subnets become the primary network-level control — ensure they are configured to restrict lateral movement between spokes
- **Private endpoint security**: Data still flows over private endpoints (no public internet exposure), maintaining data-in-transit confidentiality

Fix 2: Adjacent Private Endpoint

- **Preserves forced tunneling** — the catch-all UDR remains active, maintaining the central inspection/logging posture for all non-PE traffic
- **Storage data bypasses the gateway** via `/32 InterfaceEndpoint` routes, but stays entirely within the Azure backbone (PE-to-PE, never touches the internet)
- **Ancillary traffic still inspected**: AAD auth, DNS, management traffic continues through the gateway — compliance logging for authentication flows is maintained
- **Private Link compliance**: Data plane traffic uses Azure Private Link, which satisfies most data-sovereignty and data-in-transit requirements (encrypted within Azure backbone)
- **Additional PE surface area**: Two new private endpoints in the consumer VNet mean additional resources to govern — ensure PE approval workflows, DNS zone policies, and RBAC are applied consistently
- **Best compliance fit**: For environments requiring forced tunneling for audit/inspection purposes, adjacent PE is the **least disruptive** fix because it preserves the existing network security architecture

Summary

Concern	Broken	Fix 1 (Direct Peering)	Fix 2 (Adjacent PE)
Forced tunneling preserved	✅ Yes	❌ **Removed**	✅ Yes
Central inspection point	✅ Gateway sees all traffic	❌ Gateway bypassed	⚠️ Gateway sees ancillary traffic
Data stays on private network	✅ PE traffic	✅ PE traffic	✅ PE traffic
Gateway saturation risk	⚠️ **High**	✅ None	✅ None (storage data bypassed)
NSG as primary control	⚠️ UDR overrides	✅ Required	⚠️ UDR still active
Compliance change required	—	⚠️ May violate forced tunneling mandate	✅ Minimal

Bicep Configurations

All three states are codified as self-contained Bicep deployments:

- **`bicep/lab-current/`** — Reproduces the broken hairpin state
- **`bicep/lab-fixed-direct-peering/`** — Fix 1: Direct spoke-to-spoke peering
- **`bicep/lab-fixed-adjacent-pe/`** — Fix 2: Adjacent private endpoints in consumer VNet

Deploy any configuration with:

```
az deployment group create \
  --resource-group rg-spoke-to-spoke-lab \
  --template-file bicep/<config>/main.bicep \
  --parameters adminUsername=<user> adminPublicKey='<key>' \
               allowedSshSourceIp=<ip> storageAccountSuffix=<suffix>
```